Lecture 36

Differentiability in several variables

Uses.

Summary 5: Scalar limits and continuity are defined by error bounds; limits along a restriction agree with a full limit when it exists.
Summary 8: Elementary scalar functions satisfy product and chain rules on their stated domains.
Summary 9: The scalar mean value theorem expresses a finite increment as a derivative at an intermediate point times the input increment.
Summary 12: Euclidean norms satisfy the triangle inequality and Cauchy–Schwarz.
Summary 19: A linear map between coordinate spaces has a matrix; matrix multiplication represents composition.
Summary 32: For vector functions, convergence in Euclidean norm is equivalent to convergence of all components.

Objectives. *A function of several variables may change in many directions. Partial derivatives record coordinate changes, but differentiability requires one linear approximation valid for all small increments. We prove a continuity criterion for that approximation and the matrix chain rule. For a scalar function its derivative is represented by the gradient, which determines directional changes and normals to level curves.*

By the end of this lecture you will be able to:

- Distinguish coordinate partial derivatives from differentiability and identify the linear approximation with its Jacobian matrix when it exists.
- Prove that partial derivatives existing near a point and continuous at that point imply differentiability there.
- Prove and apply the matrix chain rule for differentiable maps between finite coordinate spaces.
- Compute directional derivatives and their extrema from a differentiable scalar function's gradient, with the zero-gradient case retained.

36.1 Neighbourhoods and limits

Let $n \geq 1$ be an integer. For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the Euclidean norm is $\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$. An *open ball* of radius $r > 0$ about a is the set $B_r(a) = \{x \in \mathbb{R}^n : \|x - a\| < r\}$. A set $U \subset \mathbb{R}^n$ is *open* when every $a \in U$ has some ball $B_r(a) \subset U$. This condition permits small increments in every direction at each point. The vectors e_j denote the standard coordinate vectors, with component 1 in position j and zero elsewhere.

Let U be open and let $F : U \to \mathbb{R}^m$, where $m \geq 1$ is an integer. A limit $F(x) \to b$ as $x \to a$ means that for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\|F(x) - b\| < \varepsilon$ whenever $x \in U$ and $0 < \|x - a\| < \delta$. Continuity at $a \in U$ means $F(x) \to F(a)$ as $x \to a$. Component limits are equivalent to this norm limit, since $|w_i| \leq \|w\| \leq \sum_i |w_i|$ for every finite vector w. The single error bound must control every allowed approach to the point. Checking a few lines can disprove a proposed limit but cannot in general prove it.

Coordinate functions are continuous because $|x_j - a_j| \leq \|x - a\|$. The scalar error-bound proofs of Lecture 5's sum, product and quotient limit laws apply with $\|x - a\|$ in place of the one-variable distance. For products the error bound is $|uv - u_0v_0| \leq |u||v - v_0| + |v_0||u - u_0|$, with u bounded near the point. For reciprocals, $v_0 \neq 0$ gives $|v| \geq |v_0|/2$ near the point and $|1/v - 1/v_0| \leq 2|v - v_0|/|v_0|^2$. Composition with a continuous scalar function also preserves continuity: first choose its input-error tolerance, then the small norm neighbourhood which makes the inner scalar value meet that tolerance. Thus continuous coordinate functions combine continuously by sums, products, quotients with nonzero denominators and continuous scalar compositions.

36.2 Partial derivatives and a candidate matrix

For a scalar function $f : U \to \mathbb{R}$, the *partial derivative* with respect to coordinate x_j at a is

$$\frac{\partial f}{\partial x_j}(a) = \lim_{t \to 0} \frac{f(a + te_j) - f(a)}{t},$$

when this real limit exists. Only the jth coordinate varies in this quotient. For a vector map $F = (F_1, \ldots, F_m)$ with all coordinate partial derivatives at a, its *Jacobian matrix* is the $m \times n$ matrix

$$J_F(a) = \left(\frac{\partial F_i}{\partial x_j}(a) \right)_{1 \leq i \leq m,\, 1 \leq j \leq n}.$$

Its row index labels an output component and its column index labels an input coordinate. It is a candidate linear approximation; existence of its entries alone does not yet justify such an approximation.

For example, the scalar function $T(x, y) = x^2 + 2xy + 3y^2$ has partial derivatives $T_x = 2x + 2y$ and $T_y = 2x + 6y$. At $P = (1, -1)$ its candidate derivative row is $(0, -4)$. We next test that linear approximation using the full definition.

36.3 One linear approximation for all increments

A map $F : U \to \mathbb{R}^m$ is *differentiable at* $a \in U$ if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$F(a + h) = F(a) + Lh + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \longrightarrow 0 \quad (h \to 0,\ h \neq 0). \tag{36.1}$$

The map L is the *derivative*, written $DF(a)$. The remainder divided by the full input length must tend to zero along all small increments, not just along coordinate axes. We also write $r(h) = o(\|h\|)$ to mean this norm-ratio condition.

Every linear map in finite coordinates is bounded by a constant times the input norm. Indeed, let its matrix entries be a_{ij}. Cauchy–Schwarz applied to each row gives

$$\|Lh\|^2 = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} h_j \right)^2 \leq \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right) \|h\|^2 .$$

Thus $\|Lh\| \le C\,\|h\|$, with the finite nonnegative constant $C = (\sum_{i,j} a_{ij}^2)^{1/2}$. This estimate will control errors in the chain rule.

The derivative is unique. If L and M both satisfy Eq. (36.1), subtract the two expansions and set $h = te_j$. The difference of their remainder quotients tends to zero, so $\|(L-M)e_j\| = 0$ for every j. Every matrix column of $L - M$ is therefore zero, hence $L = M$. Differentiability also implies continuity, because $\|F(a+h) - F(a)\| \le C\,\|h\| + \|r(h)\| \to 0$.

Setting $h = te_j$ in a differentiable map's expansion gives

$$\frac{F(a+te_j) - F(a)}{t} = Le_j + \frac{r(te_j)}{t} \longrightarrow Le_j.$$

The last error has norm tending to zero, since $\|te_j\| = |t|$. Its components are exactly the coordinate partial derivatives. Thus a differentiable map has all those partial derivatives and the matrix of $DF(a)$ is $J_F(a)$. The derivative is a linear map, while the Jacobian is its coordinate matrix.

Example 36.1 (A direct linear approximation)**.** For the scalar function $T(x,y) = x^2 + 2xy + 3y^2$, find its derivative at $P = (1,-1)$ directly from the definition. The value is $T(P) = 2$. Expanding at P gives

$$T(1+h,-1+k) = 2 - 4k + h^2 + 2hk + 3k^2.$$

The candidate linear part is $L(h,k) = -4k$. Its remainder satisfies $|h^2+2hk+3k^2| \le 2h^2+4k^2 \le 4(h^2+k^2)$, using $2|hk| \le h^2 + k^2$. Therefore the remainder divided by $\sqrt{h^2+k^2}$ is at most $4\sqrt{h^2+k^2} \to 0$. This proves differentiability and identifies $DT(P)(h,k) = -4k$. For small increments the value is approximated by $2 - 4k$, with an error bounded by four times the squared input length.

For a differentiable scalar function $f(x,y)$ at (a,b), the graph's *tangent plane* is the affine linear graph

$$z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

The derivative definition makes the vertical error between the graph and this plane $o(\sqrt{(x-a)^2+(y-b)^2})$. For the graph of T at $(1,-1,2)$, the plane is $z = 2 - 4(y+1)$. The plane represents the two-input linear approximation, rather than only one coordinate slice of the surface.

Example 36.2 (Continuity and partial derivatives do not suffice)**.** Define $f(0,0) = 0$ and $f(x,y) = x^3/(x^2+y^2)$ away from the origin. The bound $|f(x,y)| \le |x| \le \sqrt{x^2+y^2}$ proves continuity at zero. On the axes, $f(t,0) = t$ and $f(0,t) = 0$. Thus the partial derivatives at zero are $f_x = 1$ and $f_y = 0$. If the function were differentiable there, its linear part would have to be $L(h,k) = h$. But along $(h,k) = (t,t)$ with $t \neq 0$,

$$\frac{|f(t,t) - L(t,t)|}{\sqrt{t^2+t^2}} = \frac{|t/2 - t|}{\sqrt{2}|t|} = \frac{1}{2\sqrt{2}}.$$

The ratio does not tend to zero, so the continuous function is not differentiable at the origin.

36.4 Continuous partial derivatives imply differentiability

The missing condition can be supplied by continuity of the partial derivatives. In several variables, a map is called C^1 on an open set when all its component partial derivatives exist

and are continuous there. The following pointwise version needs their continuity only at the point of approximation.

Theorem 36.3 (Continuity criterion). *Let $F : U \to \mathbb{R}^m$, with $U \subset \mathbb{R}^n$ open, and let $a \in U$. Suppose all component partial derivatives exist in a neighbourhood of a and are continuous at a. Then F is differentiable at a, with derivative matrix $J_F(a)$. In particular every C^1 map is differentiable at every point of its domain.*

Figure 36.1 shows the coordinate path used in the proof for two inputs. The proof changes one input coordinate at a time.

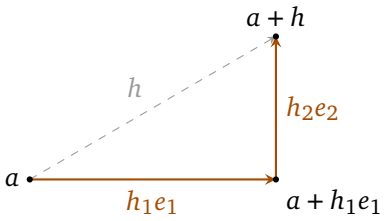


Figure 36.1: A coordinate path and its total increment in two dimensions. Positive coordinate increments are drawn; the argument also covers negative and zero increments.

Proof. Write $\partial_j F_i$ for $\partial F_i / \partial x_j$. Choose a ball about a on which all the partial derivatives exist. For sufficiently small h, all coordinate segments joining successive points a, $a + h_1 e_1$, $a + h_1 e_1 + h_2 e_2$, and so on up to $a + h$ lie in this ball. Every point on these segments is at distance at most $\|h\|$ from a. For each output component F_i, telescope its total change into the changes on these n segments. On a segment with $h_j \neq 0$, the scalar mean value theorem gives that change as $h_j \, \partial_j F_i(z_{ij}(h))$ at some intermediate point $z_{ij}(h)$. A zero coordinate increment contributes zero; in that case set $z_{ij}(h) = a$. Intermediate points may differ for different components and segments.

Subtract the candidate linear part and define its component remainder by

$$r_i(h) = \sum_{j=1}^{n} h_j \big(\partial_j F_i(z_{ij}(h)) - \partial_j F_i(a) \big).$$

All the intermediate points tend to a as $h \to 0$. There are finitely many component partial derivatives, and each is continuous at a. Hence their maximum error $\eta(h) = \max_{i,j} |\partial_j F_i(z_{ij}(h)) - \partial_j F_i(a)|$ tends to zero. Cauchy–Schwarz gives $\sum_j |h_j| \le \sqrt{n}\, \|h\|$, so

$$|r_i(h)| \le \sqrt{n}\, \eta(h)\, \|h\|, \qquad \|r(h)\| \le \sqrt{mn}\, \eta(h)\, \|h\|.$$

Dividing by $\|h\|$ gives a quantity tending to zero. This is the differentiability definition with the matrix $J_F(a)$. ∎

The criterion applies to polynomial component functions and to compositions of elementary C^1 functions, since their partial derivatives are continuous.

36.5 The matrix chain rule

Theorem 36.4 (Chain rule). *Let $F : U \to V$ and $G : V \to \mathbb{R}^p$, where $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ are open. If F is differentiable at $a \in U$ and G is differentiable at $b = F(a)$, then $G \circ F$ is differentiable at a and*

$$D(G \circ F)(a) = DG(b)\, DF(a), \qquad J_{G\circ F}(a) = J_G(F(a))J_F(a).$$

Proof. Write the two linear approximations as $F(a+h) = b + Ah + r(h)$ and $G(b+k) = G(b) + Bk + s(k)$, where $A = DF(a)$, $B = DG(b)$ and the remainder ratios tend to zero. Set $k = Ah + r(h)$. The linear-map bound and the first remainder estimate give $\|k\| \le M\,\|h\|$ for some fixed $M > 0$ and all sufficiently small h. Substitution gives

$$G(F(a+h)) = G(b) + BAh + Br(h) + s(k).$$

The term $Br(h)$ divided in norm by $\|h\|$ tends to zero by the linear-map bound. For any $\varepsilon > 0$, differentiability of G gives $\|s(k)\| \le \varepsilon\,\|k\|$ for all sufficiently small k, including $k = 0$ after setting $s(0) = 0$. Thus $\|s(k)\| \le \varepsilon M\,\|h\|$ for all sufficiently small h. As ε is arbitrary, this is also a remainder of the required order. The total linear part is BA. Matrix representation of composition gives the displayed Jacobian product, in that order. ∎

Example 36.5 (A nonlinear coordinate substitution). Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be given by $F(u,v) = (u^2, 2u+v, uv)$, and let $G : \mathbb{R}^3 \to \mathbb{R}^2$ be given by $G(x,y,z) = (xy + \sin z, x^2 + yz)$. The component partial derivatives of both maps are continuous on their full coordinate spaces by the continuity rules just proved, so the criterion gives differentiability. At $a = (1,0)$ one has $F(a) = b = (1,2,0)$. The composition $G \circ F : \mathbb{R}^2 \to \mathbb{R}^2$ has matrix

$$J_{G\circ F}(a) = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 4 & 2 \end{pmatrix}.$$

The first factor has rows $(y, x, \cos z)$ and $(2x, z, y)$ evaluated at b. The second has rows $(2u, 0)$, $(2, 1)$ and (v, u) evaluated at a. The dimensions are $(2\times 3)(3\times 2) = 2\times 2$. Direct substitution gives the same map as

$$(G \circ F)(u,v) = \bigl(u^2(2u+v) + \sin(uv),\; u^4 + (2u+v)uv\bigr).$$

Differentiating these components at $(1,0)$ gives rows $(6,2)$ and $(4,2)$, confirming the product. Reversing the matrix order would have the wrong input and output roles for this composition.

36.6 Gradients and directional changes

For a scalar differentiable function $f : U \to \mathbb{R}$, the *gradient* is the vector of its partial derivatives,

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right).$$

Its derivative acts by $Df(a)h = \nabla f(a) \cdot h$. This linear map is also called the *differential* $\mathrm{d}f_a$. Writing the coordinate increments as $\mathrm{d}x_j = h_j$ gives the familiar notation $\mathrm{d}f = \sum_j (\partial_j f)\,\mathrm{d}x_j$ for this linear approximation.

For a unit vector $u \in \mathbb{R}^n$, the *directional derivative* at a is $D_u f(a) = \lim_{t\to 0}(f(a+tu)-f(a))/t$, when the limit exists. The differentiability expansion along $h = tu$ gives

$$D_u f(a) = \nabla f(a) \cdot u.$$

The remainder divided by t tends to zero because $\|tu\| = |t|$. Cauchy–Schwarz now gives $|D_u f(a)| \le \|\nabla f(a)\|$. If the gradient is nonzero, the maximum is its norm and is attained in the direction $u = \nabla f(a)/\|\nabla f(a)\|$. The negative of that direction attains the minimum, the negative norm. If the gradient is zero, every directional derivative is zero and no unique maximising direction is selected.

For the quadratic function T, the gradient is $\nabla T(x, y) = (2x + 2y, 2x + 6y)$. At $P = (1, -1)$ it is $(0, -4)$, agreeing with the direct differential $-4\mathrm{d}y$. In the unit direction $(3/5, 4/5)$ its rate of change is $-16/5$. Its greatest directional increase is 4, in direction $(0, -1)$.

A *level set* is the set where f equals a fixed constant. If a differentiable curve $r(t)$ lies in such a set, then the chain rule gives $0 = (f(r(t)))' = \nabla f(r(t)) \cdot r'(t)$. Where both vectors are nonzero, the gradient is perpendicular to the curve's tangent direction. This statement assumes the curve is given; Lecture 37 states hypotheses that locally produce level curves. For $T = 2$, one branch near P is explicitly $y = (-x - \sqrt{6 - 2x^2})/3$. Substitution gives $T = 2$, and differentiation at $x = 1$ gives $y' = 0$. The resulting regular graph curve has a horizontal tangent there.

Figure 36.2 shows this level curve and its gradient direction at P.

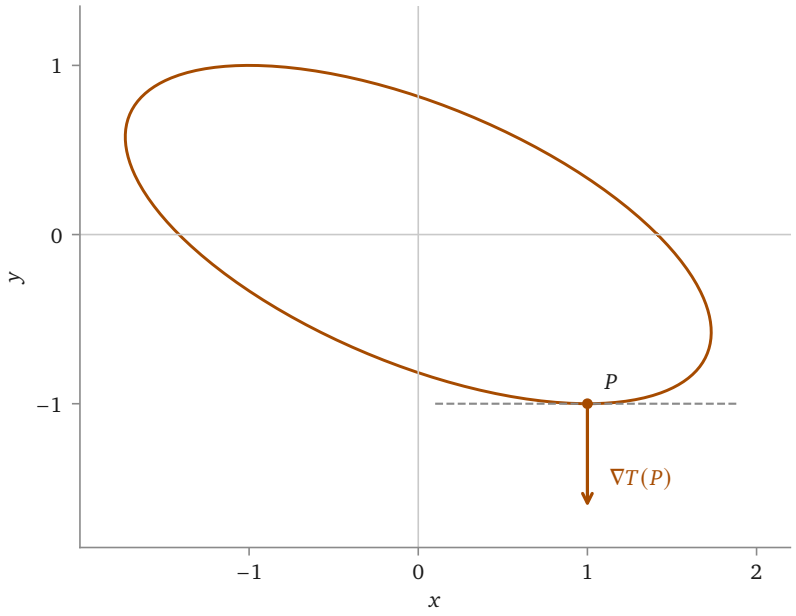


Figure 36.2: The level curve $T(x, y) = 2$, the point $P = (1, -1)$ and its horizontal tangent. The gradient arrow points downward; its drawn length indicates direction rather than the gradient magnitude.

Summary 36.

- Differentiability means the single linear expansion Eq. (36.1) with error $o(\|h\|)$ along every small increment. The derivative is unique, implies continuity and has Jacobian matrix $J_F(a)$.
- Partial derivatives, even with continuity of the function, do not suffice. Theorem 36.3 supplies a sufficient condition: partial derivatives exist near the point and are continuous at the point.
- Theorem 36.4 gives the Jacobian of a composition as the outer Jacobian evaluated at the inner image times the inner Jacobian.
- For a differentiable scalar function, $D_u f = \nabla f \cdot u$ for unit u. The maximum is $\|\nabla f\|$; a zero gradient makes all these rates zero. A nonzero gradient is perpendicular to any given regular curve in its level set.

Sheet 36 tests the full remainder, coordinate calculations and gradient interpretation. Lecture 37 uses derivatives to describe local equations and to classify critical points.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§14.4–14.6. W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw–Hill, 1976, Chapter 9 (derivatives in several variables).